CONNECTING 3 VPC's WITH TRANSIT GATEWAY

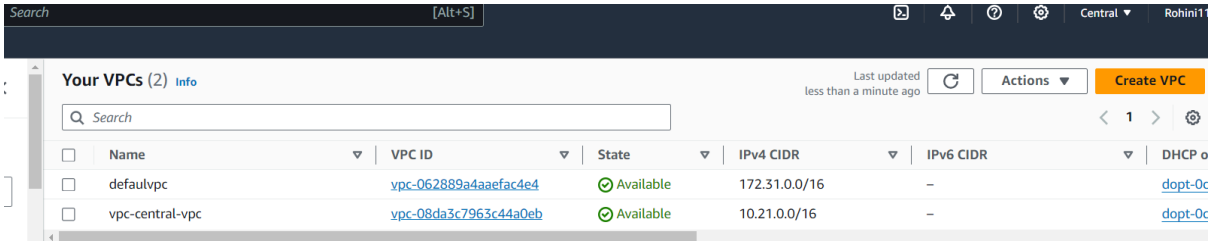
Transit Gateway

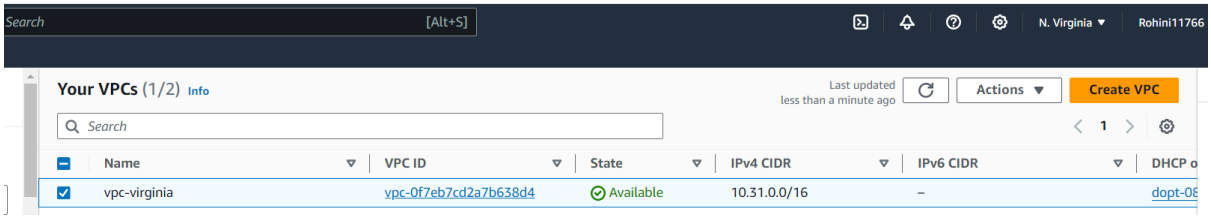
A transit gateway is a network transit hub that you can use to interconnect your virtual private clouds (VPCs) and on-premises networks. As your cloud infrastructure expands globally, inter-Region peering connects transit gateways together using the AWS Global Infrastructure. All network traffic between AWS data centers is automatically encrypted at the physical layer.

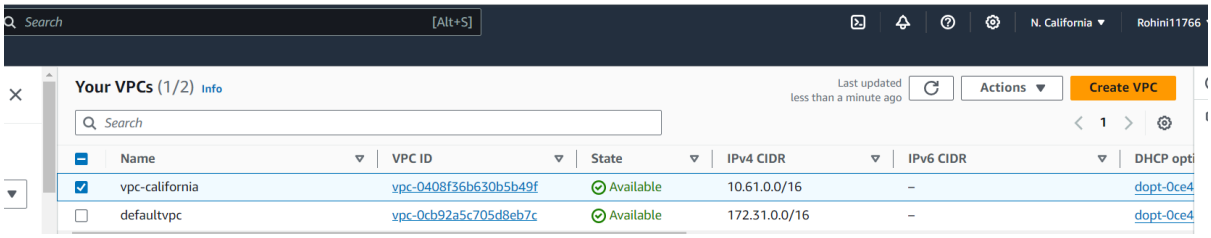
- ➢ Creating the transit gateway btw 3 vpc in different regions

1. Vpc-central -10.21.0.0/16
2. Vpc – Virginia -10.31.0.0/16
3. Vpc – California -10.61.0.0/16

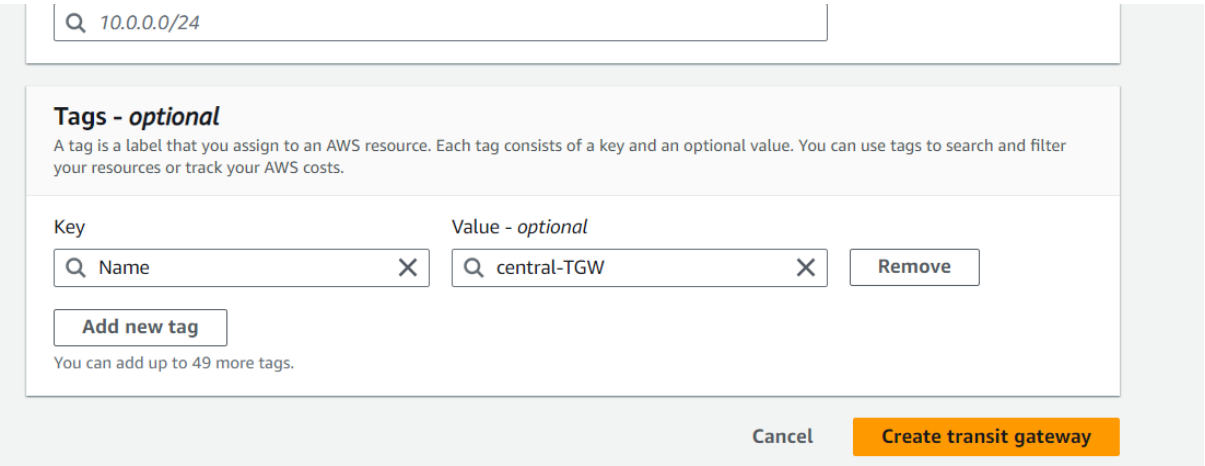
Created the three vpc in three different regions

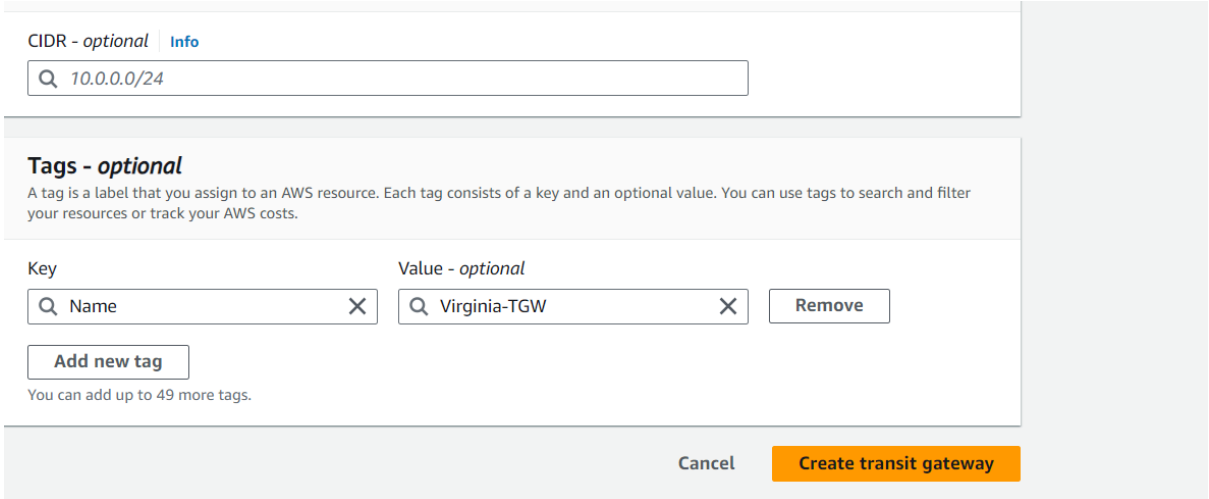


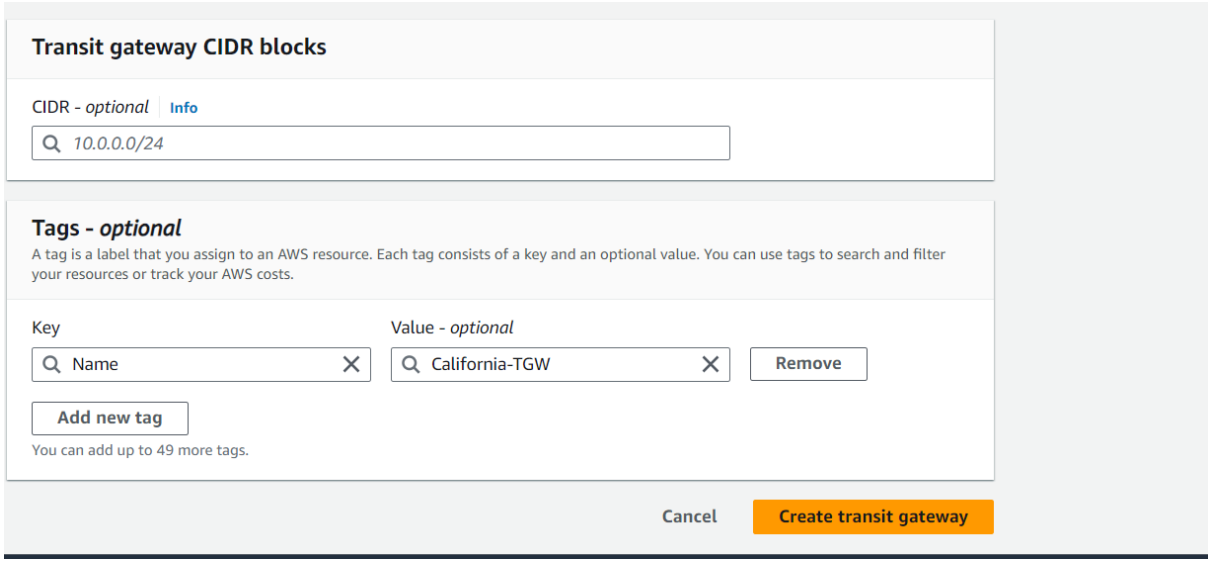


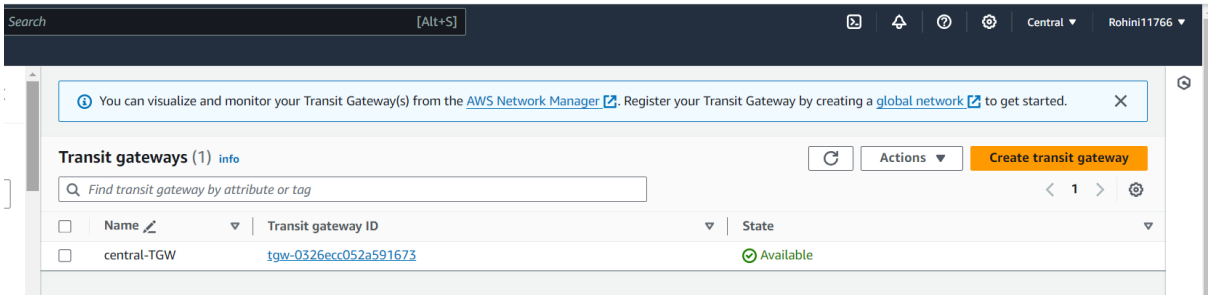


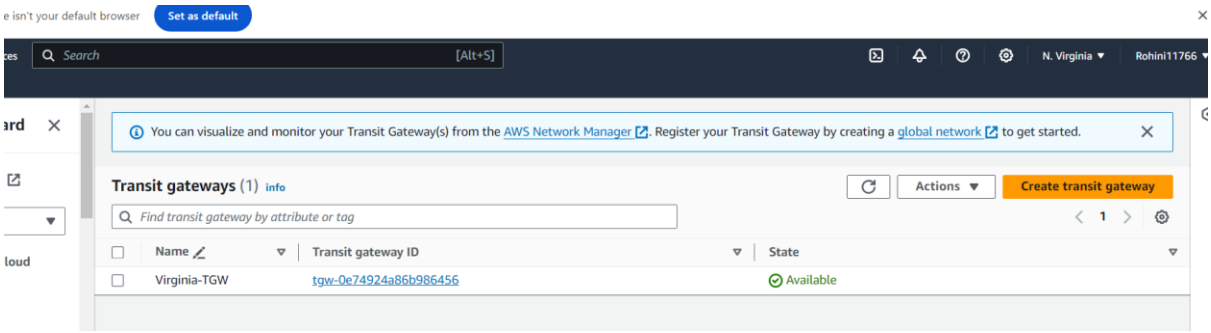
➢ NOW CREATING THE 3 TRANSIT GATEWAY IN 3 DIFFERENT ZONES

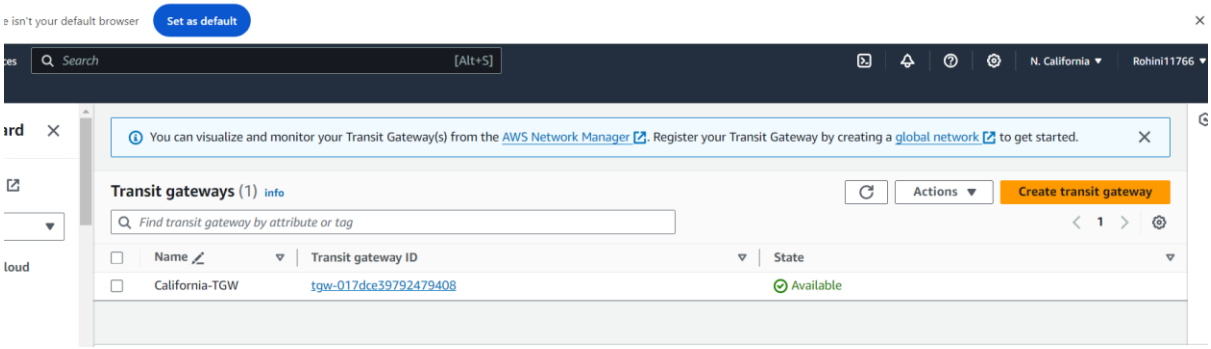








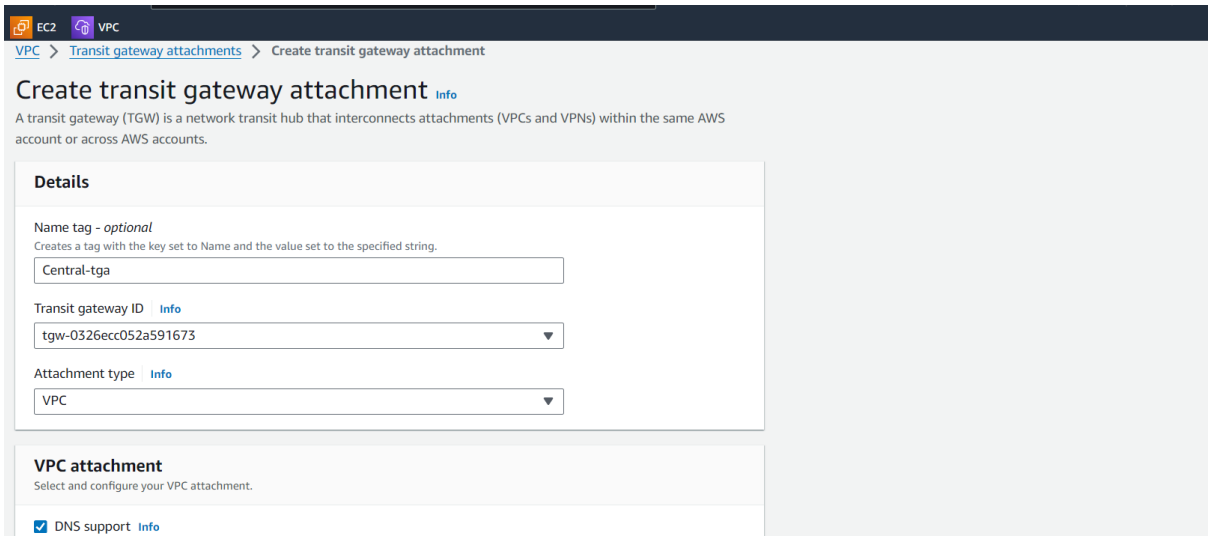


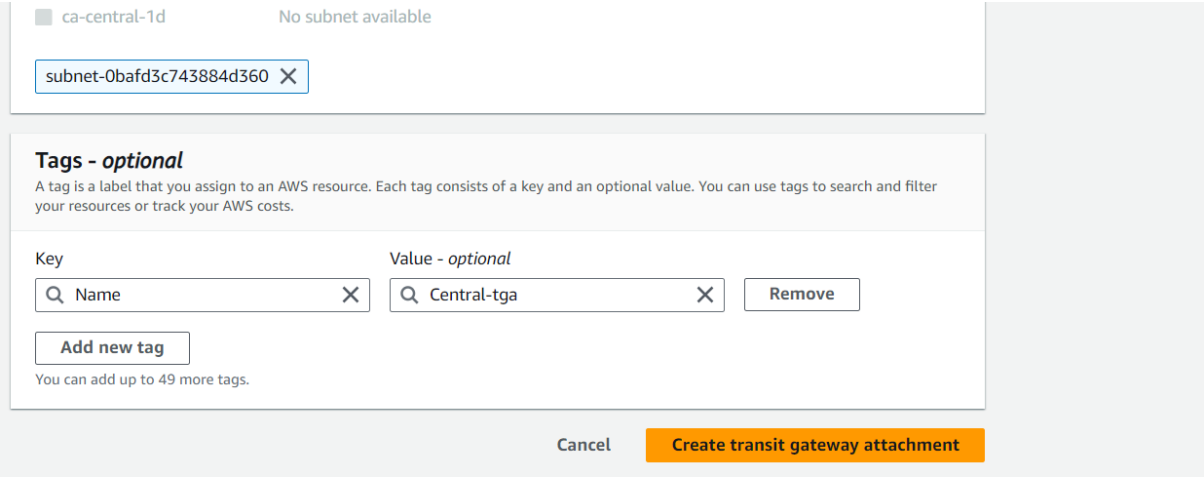


Created the transit gateway

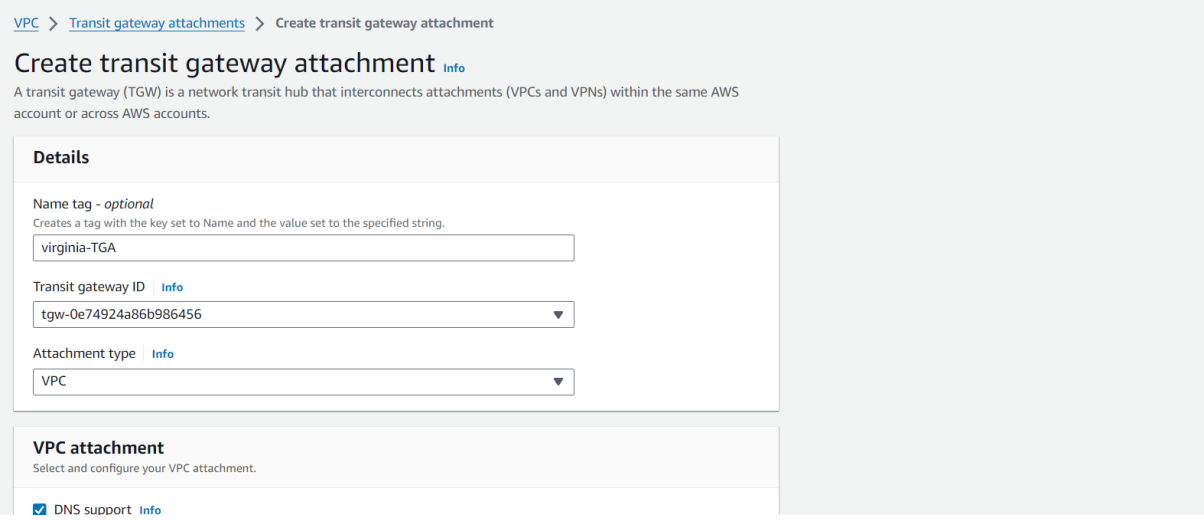
- Now creating the transit attachment in three regions

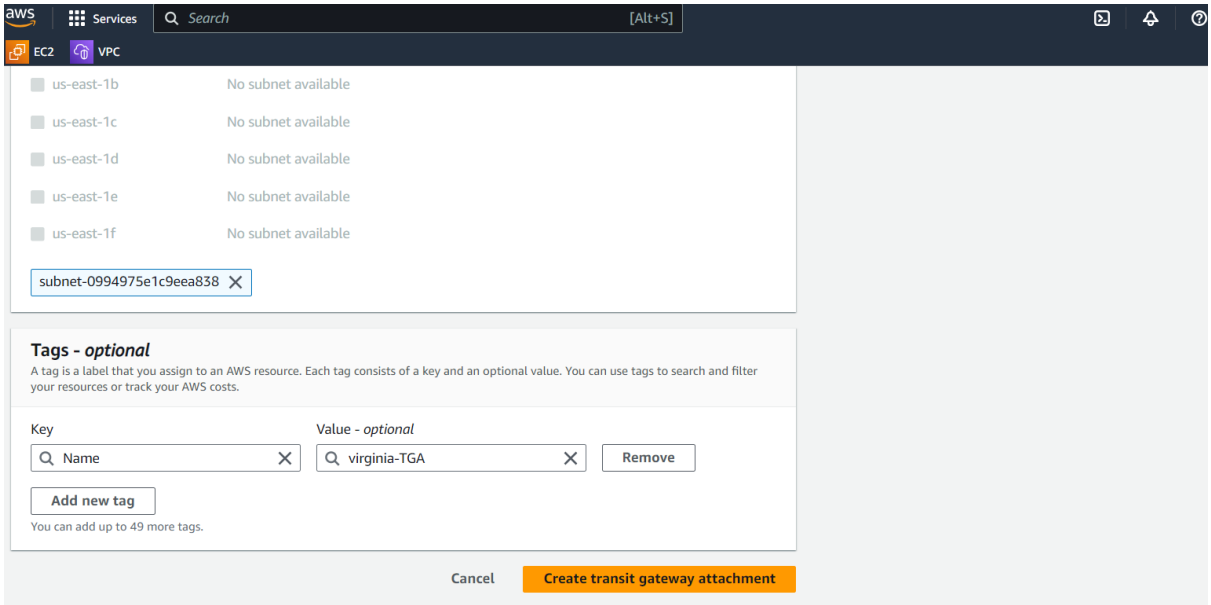
CENTRAL



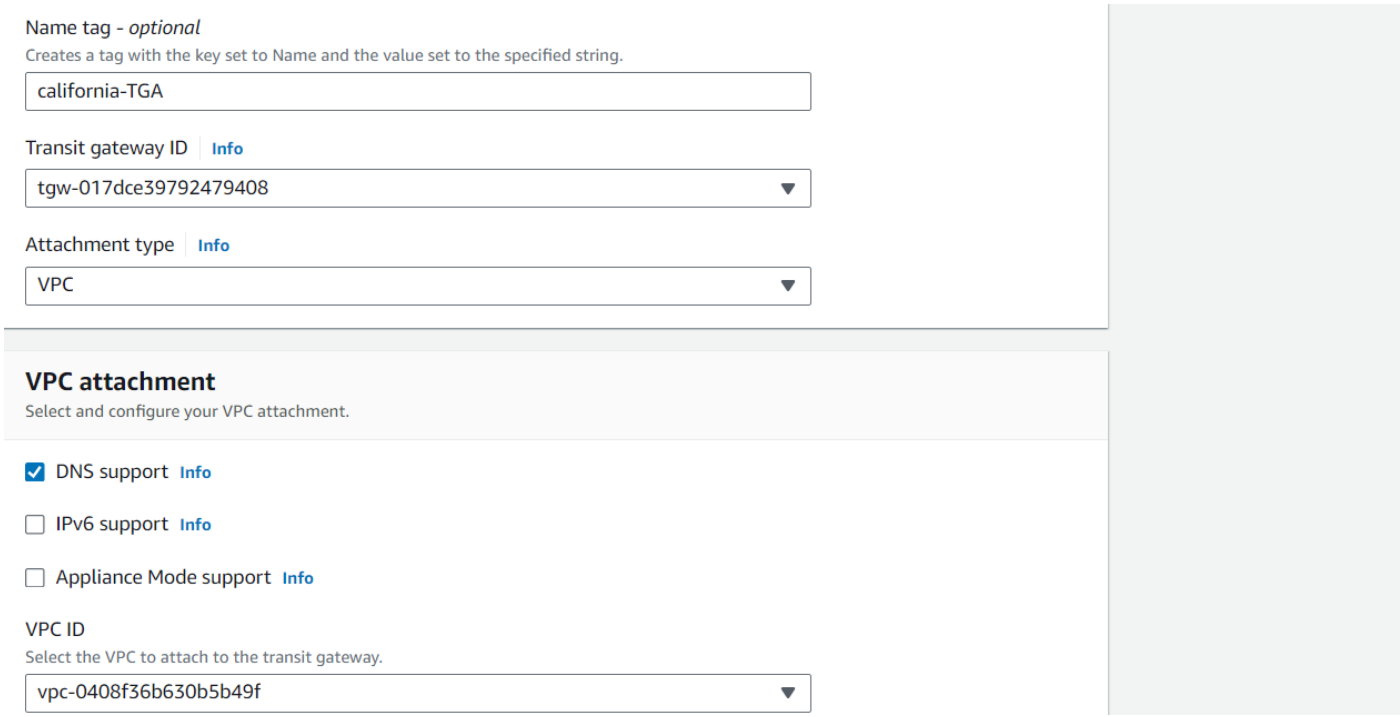

ca-central-1d
No subnet available
subnet-0bafd3c743884d360
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
Central-tga
Remove
Add new tag
You can add up to 49 more tags.
Cancel
Create transit gateway attachment

Virginia


VPC > Transit gateway attachments > Create transit gateway attachment
Create transit gateway attachment Info
A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.
Details
Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.
virginia-TGA
Transit gateway ID Info
tgw-0e74924a86b986456
Attachment type Info
VPC
VPC attachment
Select and configure your VPC attachment.
DNS support Info



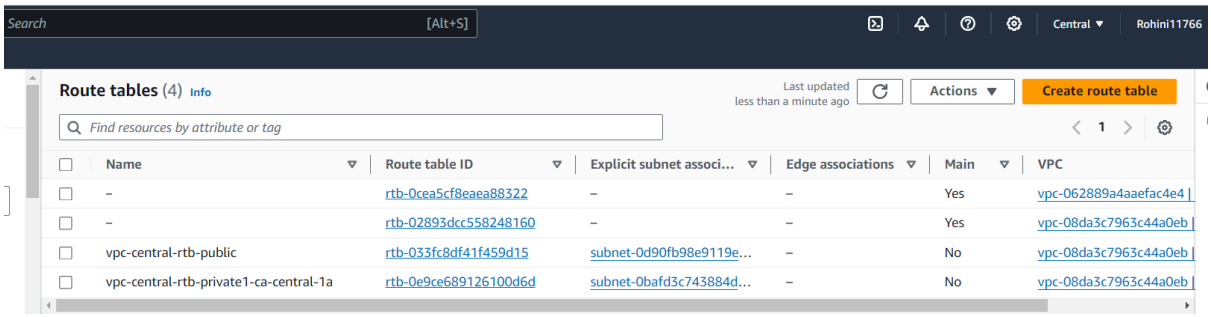
California

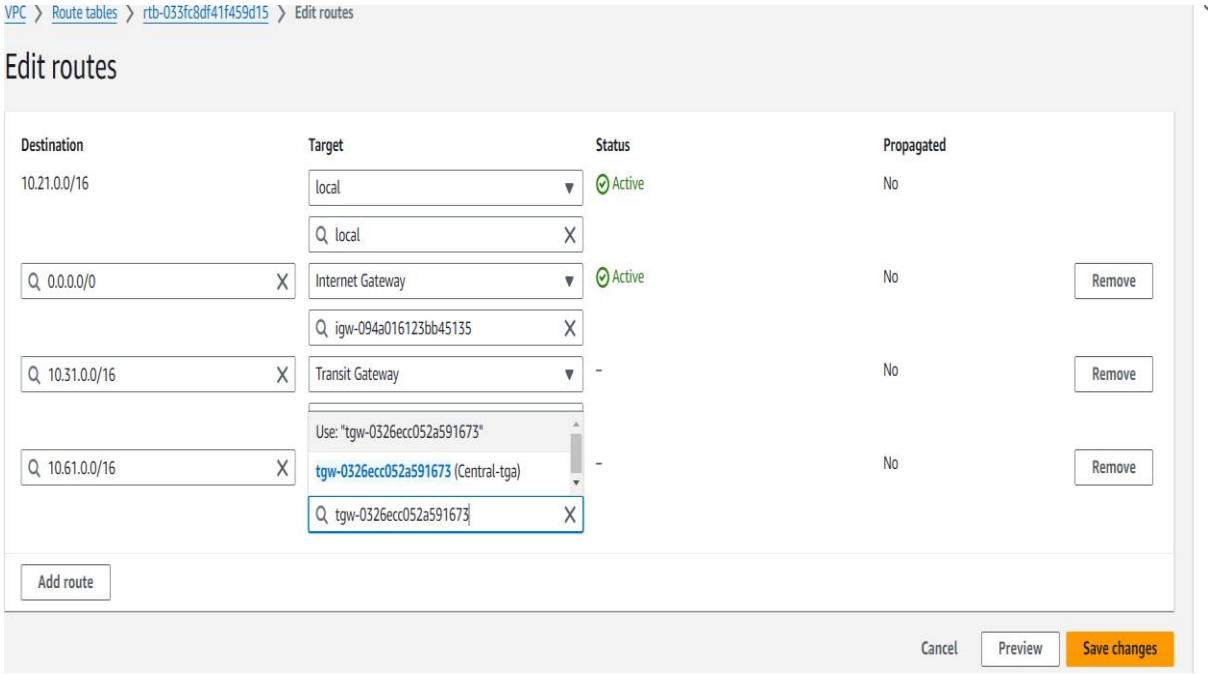


Now gng to route tables of each vpc and connecting the Transit gateway by giving the cidr as the two other vpc one

1. Vpc-central -10.21.0.0/16
2. Vpc – Virginia -10.31.0.0/16
3. Vpc – California -10.61.0.0/16

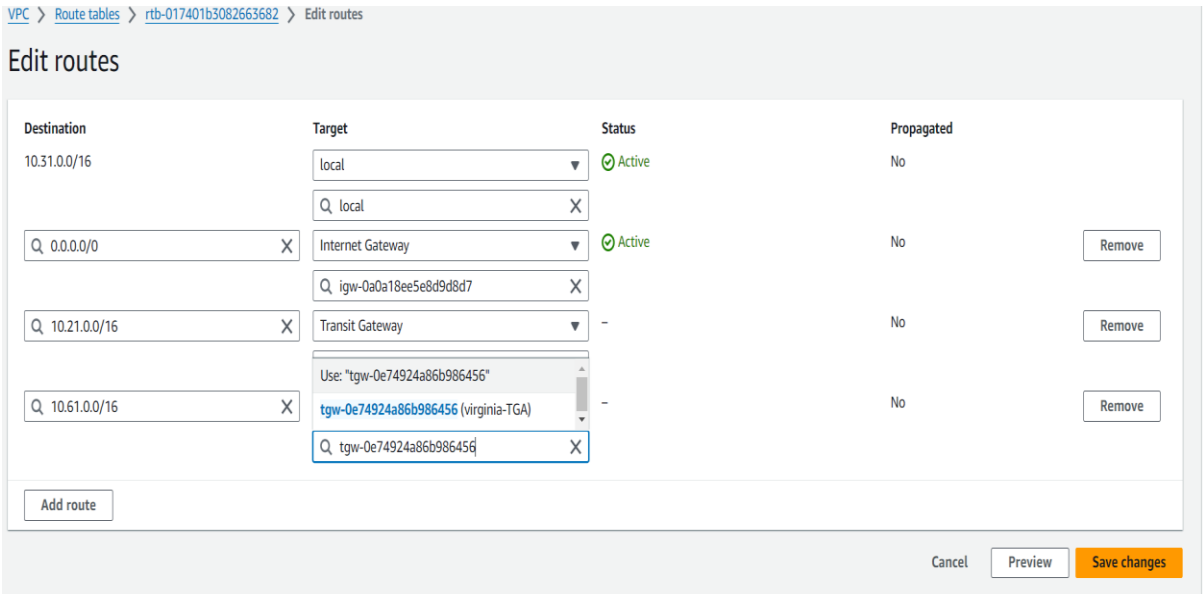
Changing in the vpc-central





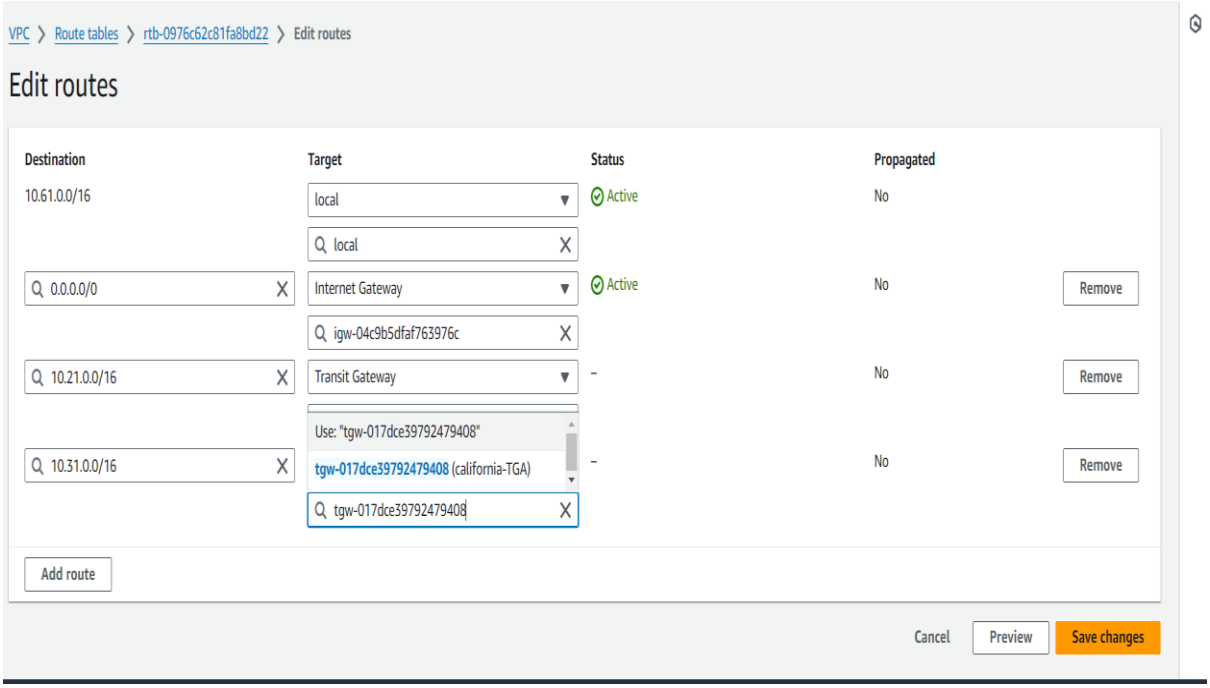
Saving the changes

For vpc – Virginia



Saving the changes

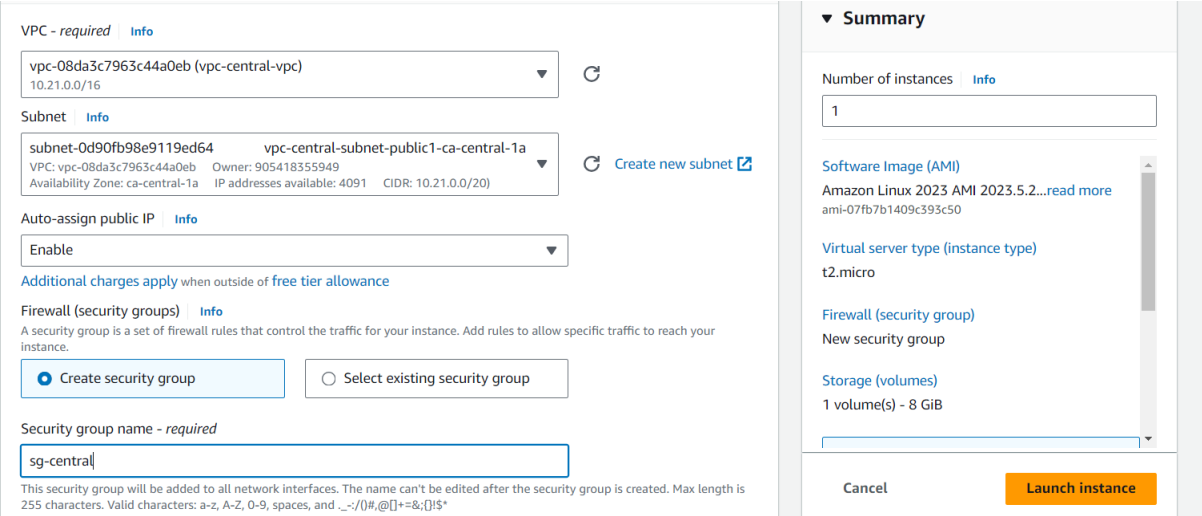
For California



Saving the changes

- Now we need to launch the instances in 3 different regions

Instance-central

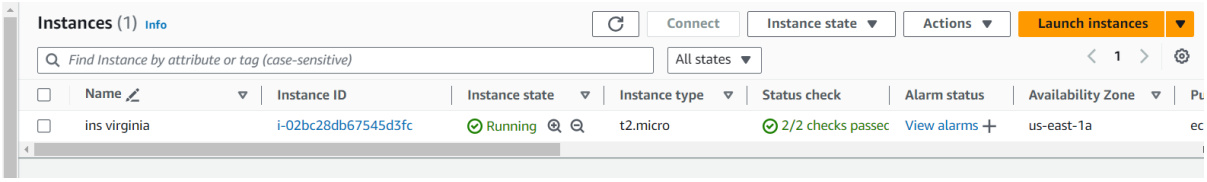


```
[root@ip-10-21-4-98 ~]# cd /usr/share/nginx/html
[root@ip-10-21-4-98 html]# cd
[root@ip-10-21-4-98 ~]# ls
[root@ip-10-21-4-98 ~]# cd /usr/share/nginx/html
[root@ip-10-21-4-98 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-10-21-4-98 html]# rm index.html
rm: remove regular file 'index.html'? yes
[root@ip-10-21-4-98 html]# vi index.html
[root@ip-10-21-4-98 html]# systemctl restart nginx
[root@ip-10-21-4-98 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-10-21-4-98 html]#
```

- Connected with private ip

```
[root@ip-10-21-4-98 html]# curl 10.21.4.98:80
This is from central
[root@ip-10-21-4-98 html]#
```

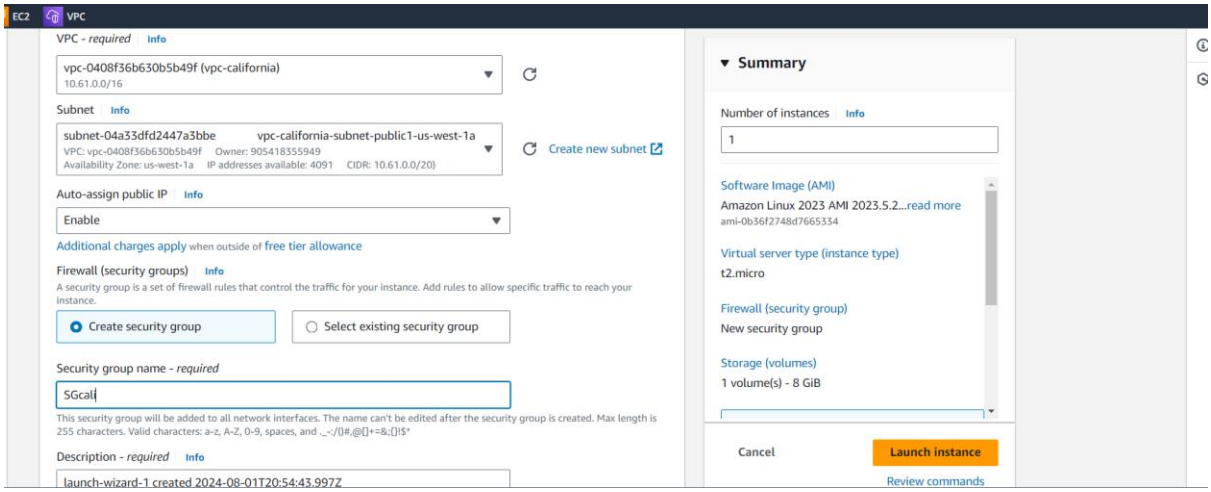
- Instance is created in Virginia



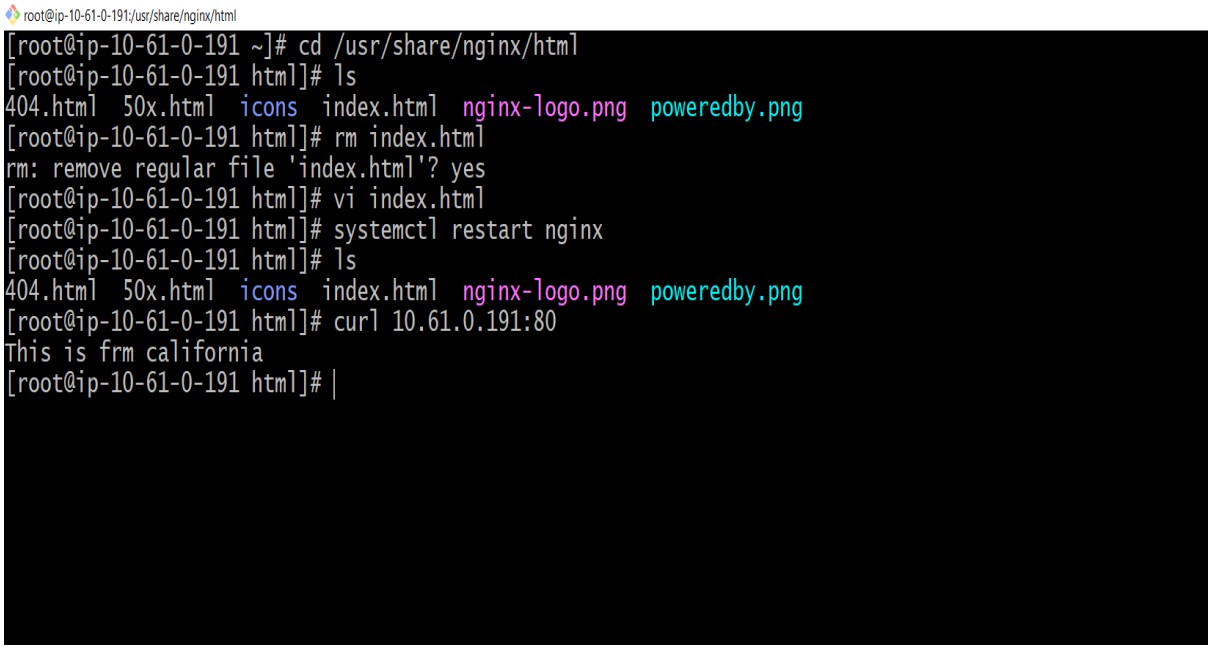
root@ip-10-31-8-129:/usr/share/nginx/html

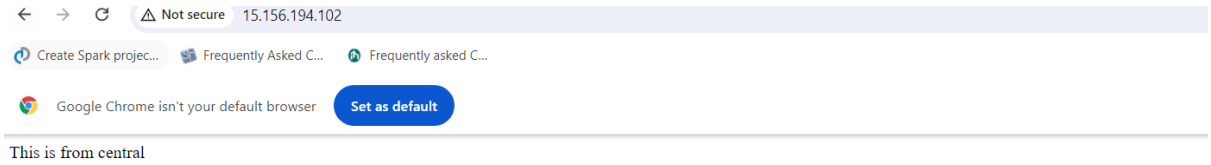
```
[root@ip-10-31-8-129 ~]# cd /usr/share/nginx/html
[root@ip-10-31-8-129 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-10-31-8-129 html]# rm index.html
rm: remove regular file 'index.html'? yes
[root@ip-10-31-8-129 html]# vi index.html
[root@ip-10-31-8-129 html]# systemctl restart nginx
[root@ip-10-31-8-129 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-10-31-8-129 html]# curl 10.31.8.129:80
This is from virginia
[root@ip-10-31-8-129 html]#
```

- Instance in California

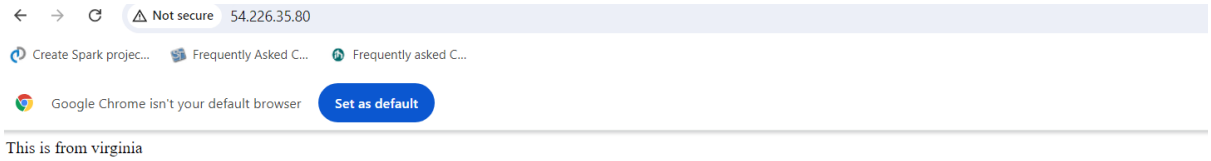


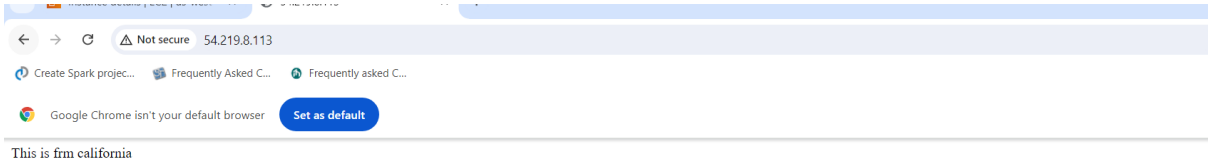
- Launching the instance
- Connected with the instance
- And we are able to connect with the private ip also



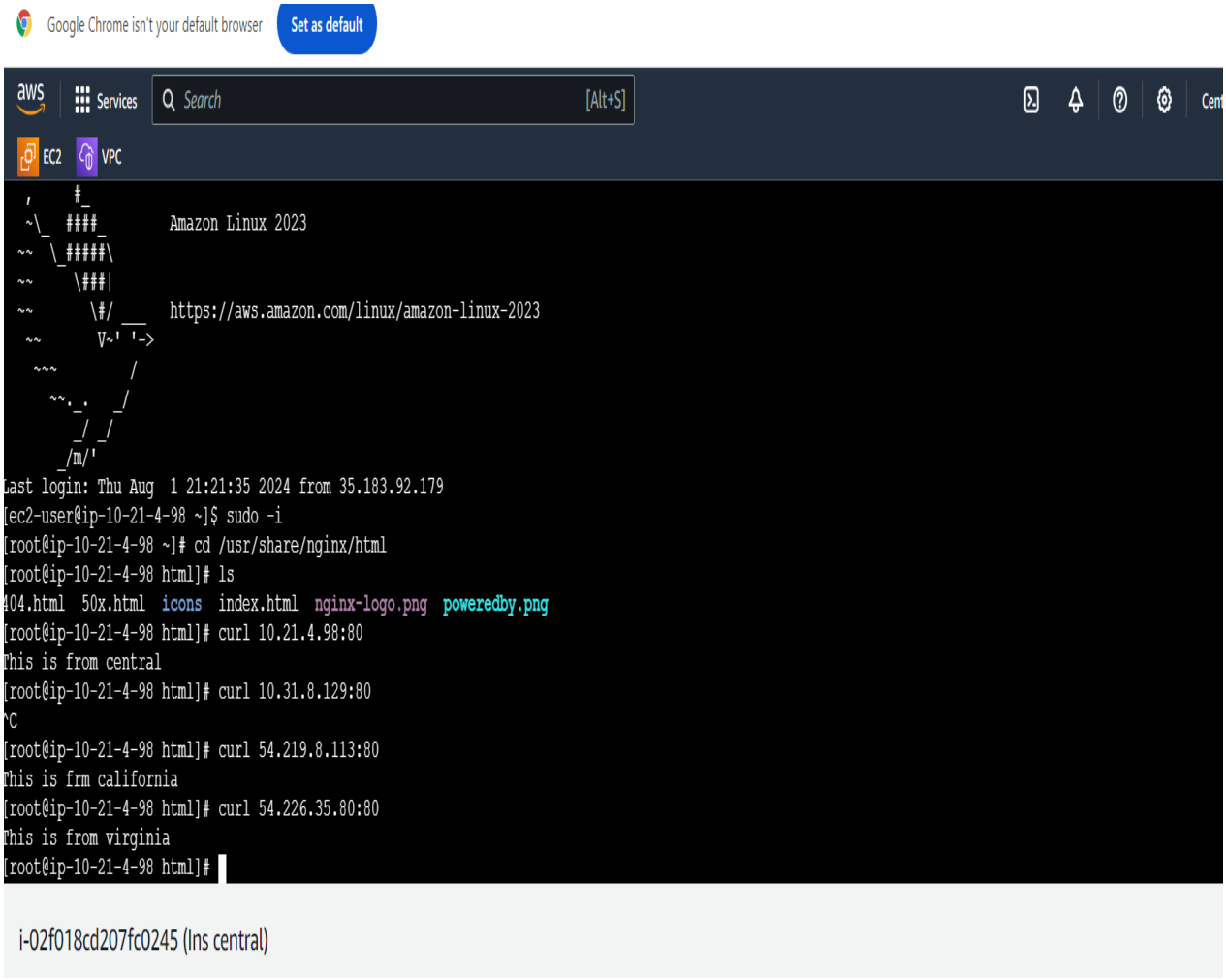


This is from central





- We are connecting frm one region instance to the different regions instances



We can see it is reachable. This is how the Transit gateway works...!!!!

- Conclusion:

AWS Transit Gateway simplifies cloud network architectures by acting as a hub to interconnect your VPCs, VPNs, and data centers. It eliminates complex mesh topologies and provides easy scalability, centralized management, and secure network segmentation. As your cloud footprint grows, Transit Gateway is key to maintaining a simple.